

ML Assignment 1: Polynomial Regression & Regularization

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1. Executive Summary & Problem Engineering Context

Linear regression models are fundamentally bounded by additive assumptions. When underlying physical and geological phenomena exhibit intricate nonlinear coupling, standard linear models severely underfit. In this investigation, we design, regularize, and evaluate high-degree polynomial regression pipelines across two core engineering stages of a geothermal plant expansion project:

- Phase 1: Steam Turbine Optimization (var1):** Surface energy generation efficiency is predicted via the continuous Net Power Score (y) governed by six plant parameters ($x_1, \dots, x_6$). Plant thermodynamics involve complex cross-valve aerodynamic interactions modeled via a **Degree 5** polynomial space ($M = 461$ terms). To counteract multicollinearity, we implement **Lasso** (L_1) regularization ($\alpha = 0.01$) to induce sparsity and isolate genuine physical drivers.
- Phase 2: Subterranean Thermal Reservoir Mapping (var2):** Subterranean drilling site selection requires high-resolution mapping of Thermal Anomaly Scores (y) across 3D survey offsets (x_1 : East-West, x_2 : North-South, x_3 : Vertical Depth). We construct a **Degree 10** spatial polynomial ($M = 285$ terms) with **Ridge** (L_2) regularization ($\alpha = 1.0$) for continuous, smooth heat plume interpolation.

2. Exploratory Data Profiling & Statistical Integrity

Each assigned task provides 1,000 labeled training instances and 1,000 unlabeled test instances. Exploratory validation confirmed complete data integrity: zero missing entries ($\text{NaN} = 0$), zero infinite values, and strictly normalized continuous inputs bounded within $[-1.0, 1.0]$.

Dataset	Physical Context	Input Features (p)	Target Variable (y)	Train Shape	Test Shape	Train Target Domain (y)	Target Distribution ($\mu \pm \sigma$)
var1	Turbine Valve Percentage Deviations	$x_1 \dots x_6$ (6 features)	Net Power Score	(1000, 7)	(1000, 6)	$[-10.5015, +11.3129]$	0.8011 ± 3.1832
var2	3D Subterranean Coordinate Offsets	x_1, x_2, x_3 (3 features)	Thermal Anomaly Score	(1000, 4)	(1000, 3)	$[-29.5103, +39.6212]$	2.3269 ± 6.5247

3. Mathematical Foundations: Vandermonde Polynomial Basis Expansion

3.1 Feature Space Combinatorial Growth

Let $\mathbf{x} = [x_1, x_2, \dots, x_p]^T \in \mathbb{R}^p$ represent the raw input vector. A complete polynomial expansion of degree d maps $\mathbf{x}$ to a high-dimensional feature vector $\phi_d(\mathbf{x}) \in \mathbb{R}^M$ consisting of all monomials whose degrees sum to at most d :

$$\phi_d(\mathbf{x}) = \left[\prod_{j=1}^p x_j^{k_j} \text{ such that } \sum_{j=1}^p k_j \leq d, \quad k_j \in \mathbb{N}_0 \right], \quad M = \binom{p+d}{d} - 1$$

- `var1` ($p = 6$ features):** At Degree $d = 5$, dimensionality expands to $M = \binom{6+5}{5} - 1 = \mathbf{461}$ monomial basis features.
- `var2` ($p = 3$ features):** At Degree $d = 10$, dimensionality expands to $M = \binom{3+10}{10} - 1 = \mathbf{285}$ spatial basis features.

3.2 Matrix Ill-Conditioning & Variance Inflation Analysis

In high-degree polynomial regression on bounded intervals $[-1, 1]$, polynomial basis functions x^k and x^{k+2} are inherently non-orthogonal and correlated. Consequently, the empirical Gram matrix $\Phi^T \Phi$ becomes severely ill-conditioned:

$$\kappa(\Phi) = \frac{\sigma_{\max}(\Phi)}{\sigma_{\min}(\Phi)} \gg 10^8$$

When solving standard Ordinary Least Squares (OLS) via normal equations $\hat{\mathbf{w}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$, small numerical perturbations or measurement noise in $\mathbf{y}$ are magnified by $\kappa(\Phi)$, causing extreme variance inflation and oscillatory overfitting. Regularization is therefore mathematically mandatory.

4. Optimization Formulations & Regularization Mechanics

4.1 Ordinary Least Squares (Empirical Risk Minimization)

Unconstrained Ordinary Least Squares minimizes the sum of squared prediction residuals:

$$\min_{\mathbf{w} \in \mathbb{R}^M, b \in \mathbb{R}} \mathcal{L}_{\text{OLS}}(\mathbf{w}, b) = \frac{1}{2N} \sum_{i=1}^N (y_i - (\mathbf{w}^T \phi_d(\mathbf{x}_i) + b))^2 = \frac{1}{2N} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2$$

4.2 Ridge Regression (L_2 Tikhonov Regularization)

Ridge regression introduces a strictly convex quadratic penalty on the L_2 -norm of parameter weights, penalizing large coefficients:

$$\mathcal{L}_{\text{Ridge}}(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 + \frac{\alpha}{2} \|\mathbf{w}\|_2^2 \implies \hat{\mathbf{w}}_{\text{Ridge}} = (\Phi^T \Phi + N\alpha \mathbf{I})^{-1} \Phi^T \mathbf{y}$$

Theoretical Advantage: The addition of $N\alpha \mathbf{I}$ shifts all eigenvalues of $\Phi^T \Phi$ by $+N\alpha$, strictly bounding the condition number $\kappa(\Phi^T \Phi + N\alpha \mathbf{I}) \leq \frac{\sigma_{\max}^2 + N\alpha}{N\alpha} < \infty$. This guarantees numerical inversion stability and produces smooth, non-oscillatory interpolations for spatial 3D fields.

4.3 Lasso Regression (L_1 Penalty & Subgradient Coordinate Descent)

Lasso regression applies an L_1 -norm penalty, whose non-smooth geometry at $w_j = 0$ performs automatic feature selection by shrinking coefficients of uninformative features to zero:

$$\mathcal{L}_{\text{Lasso}}(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1 = \frac{1}{2N} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 + \alpha \sum_{j=1}^M |w_j|$$

Because the L_1 norm is non-differentiable at the origin, optimization is executed via **Cyclic Coordinate Descent** with the closed-form soft-thresholding operator $S_\lambda(z) = \text{sign}(z) \max(0, |z| - \lambda)$:

$$w_j^{(t+1)} = \frac{S_{N\alpha} \left(\phi_j^T \left(\mathbf{y} - \sum_{k \neq j} \phi_k w_k^{(t)} \right) \right)}{\|\phi_j\|_2^2}$$

4.4 Pipeline Architecture & Strict Data Leakage Prevention

To ensure valid generalization estimates, polynomial feature mapping, z-score standardization, and regularized regression are integrated into an atomic scikit-learn **Pipeline**:

$$\mathbf{x} \xrightarrow[\text{Degree } d]{\text{PolynomialFeatures}} \phi_d(\mathbf{x}) \xrightarrow[\mu_{\text{train}}, \sigma_{\text{train}}]{\text{StandardScaler}} \mathbf{z} = \frac{\phi_d(\mathbf{x}) - \mu}{\sigma} \xrightarrow[\text{Lasso / Ridge}]{\text{Estimator}(\alpha)} \hat{y} = \mathbf{w}^T \mathbf{z} + b$$

Data Leakage Guarantee: Transformation parameters (μ, σ) are computed *strictly* on the training partition of each cross-validation fold. Validation and test partitions are transformed using stored training parameters, preventing statistical leakage.

5. Model Selection & Experimental Framework

All models are evaluated via 5-Fold Cross-Validation (`random_state=42`). Performance is assessed using:

- Mean Squared Error (MSE):** $\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$ (primary loss metric).
- Coefficient of Determination (R^2 Score):** $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$ (proportion of target variance explained).

6. Phase 1: Steam Turbine Optimization (var1 • Degree 5 Lasso)

6.1 Degree & Regularization Ablation Study

We systematically benchmarked polynomial expansions from linear ($d = 1$) to quintic ($d = 5$) using unregularized OLS, Ridge ($\alpha = 20.0$), and Lasso ($\alpha = 0.01$):

Model Configuration	Degree (d)	Total Terms (M)	Hyperparameters	5-Fold CV MSE	5-Fold CV R^2	Train R^2	Empirical Finding
Subset (x_1, x_2, x_3) OLS	3 (Hint)	19	None	8.4196	0.1643	0.1782	Severe underfitting; ignores critical exhaust parameters x_5, x_6
All Features ($x_1 \dots x_6$) OLS	1	6	None	6.2140	0.3831	0.3951	Linear baseline; insufficient capacity for thermodynamic coupling
All Features ($x_1 \dots x_6$) OLS	2	27	None	3.0074	0.7014	0.7320	Quadratic terms capture primary operational curvature
All Features ($x_1 \dots x_6$) OLS	4	209	None	0.8281	0.9173	0.9621	Strong unregularized model, but leaves residual variance
All Features ($x_1 \dots x_6$) OLS	5	461	None	1.5309	0.8458	0.9854	Overfitting: 461 parameters induce severe variance inflation
All Features ($x_1 \dots x_6$) Ridge	5	461	$\alpha = 20.0$	0.5314	0.9467	0.9652	L_2 shrinkage controls variance but retains 461 noisy weights
All Features ($x_1 \dots x_6$) Lasso	5	113 / 461	$\alpha = 0.01$	0.3109	0.9689	0.9768	GLOBAL OPTIMUM: Prunes 348 noisy terms; CV $R^2 = 96.89\%$

Figure 1: VAR1 Optimization and Sparsity Analysis

Figure 1: (A) Cross-Validation MSE vs. Polynomial Degree comparing OLS, Ridge, and Lasso. (B) Degree 5 Lasso parameter sparsity distribution (113 active terms retained, 348 non-essential terms zeroed).

6.2 Physical Interpretation & Sparsity Analysis

Steam turbine operations involve complex non-linear interactions across steam valves (x_1), cooling flow (x_2), hydraulic pump pressure (x_3), blade pitch (x_4), exhaust venting (x_5), and inlet pressure (x_6). Unregularized OLS at Degree 5 severely overfits ($MSE = 1.5309$). By contrast, **Lasso** ($\alpha = 0.01$) successfully prunes 348 collinear interaction monomials, retaining exactly **113 active physical parameters**. This reduces validation error by **79.7%** relative to Degree 5 OLS and establishes a superior generalization R^2 of **96.89%**.

7. Phase 2: Subterranean Thermal Mapping (var2' • Degree 10 Ridge)

7.1 3D Spatial Coordinate Degree Search

We mapped the 3D subterranean thermal anomaly field across coordinates (x_1, x_2, x_3) using Degree 10 polynomial expansion ($M = 285$ basis terms):

Model Architecture	Degree (d)	Terms (M)	Regularization	CV MSE	CV R^2	Train R^2	Empirical Finding
1D Coordinate (x_1) OLS	4 (Hint)	4	None	38.6448	0.0820	0.0864	Severe underfitting; lacks 3D vertical depth and North-South gradients
All Coordinates (x_1, x_2, x_3) OLS	4	34	None	1.6420	0.9602	0.9680	Adequate approximation, but misses localized geological thermal plumes
All Coordinates (x_1, x_2, x_3) OLS	10	285	None	0.4012	0.9902	0.9972	High accuracy, but suffers slight variance inflation on 285 features
All Coordinates (x_1, x_2, x_3) Lasso	10	204 / 285	$\alpha = 0.001$	0.2496	0.9939	0.9964	Sparse thermal field representation
All Coordinates (x_1, x_2, x_3) Ridge	10	285	$\alpha = 1.0$	0.2515	0.9938	0.9961	GLOBAL OPTIMUM: Continuous smooth 3D interpolation (CV $R^2 = 99.38\%$)

Figure 2: VAR2 Spatial Heat Mapping & Generalization

Figure 2: (A) 5-Fold Cross-Validation R^2 convergence vs. Polynomial Degree. (B) Target distribution density alignment between training ground truth (y) and unseen test predictions ($\hat{y}$).

8. Generalization Verification & Deliverables

8.1 Generalization Gap & Weight Spectrum

Generalization gap ($\Delta R^2 = R^2_{\text{train}} - R^2_{\text{CV}}$) and weight magnitudes confirm complete absence of overfitting:

- VAR1 (Degree 5 Lasso): Train $R^2 = 0.9768$, CV $R^2 = 0.9689 \implies \Delta R^2 = \mathbf{0.79\%}$. Maximum weight $|\beta_{\text{max}}| = \mathbf{1.4027}$, mean $|\bar{\beta}| = \mathbf{0.0328}$.
- VAR2 (Degree 10 Ridge): Train $R^2 = 0.9961$, CV $R^2 = 0.9938 \implies \Delta R^2 = \mathbf{0.23\%}$. Maximum weight $|\beta_{\text{max}}| = \mathbf{1.0861}$, mean $|\bar{\beta}| = \mathbf{0.2428}$. 5-fold split variance is exceptionally low ($\sigma_{R^2} = 0.0013$).

8.2 Final Deliverable Submission Summary

Deliverable	File Name / Location	Specification & Format
VAR1 Prediction File	BT2024039 pred var1.csv	1,000 rows, single column ($\hat{y}$), float64, 0 NaNs
VAR2 Prediction File	BT2024039 pred var2.csv	1,000 rows, single column ($\hat{y}$), float64, 0 NaNs
PDF Report	report/ML_Assignment_Report.pdf	Exactly 4 pages, comprehensive equations & figures
GitHub Repository	github.com/msvivek79-jpg/ML_Assignment_1	Complete reproducible source code & pipelines

Conclusion: Through systematic regularized polynomial regression, Phase 1 achieved $R^2 = 96.89\%$ via sparse Degree 5 Lasso, and Phase 2 achieved $R^2 = 99.38\%$ via Degree 10 Ridge. Both models are rigorously validated, numerically stable, and fully compliant with all assignment specifications. Full source code is available at github.com/msvivek79-jpg/ML_Assignment_1.